
Machine Learning Outlier Detection for Robust Causal Inference

How outliers bias treatment effects, and a practical fix

Adrian Enders | Amazon

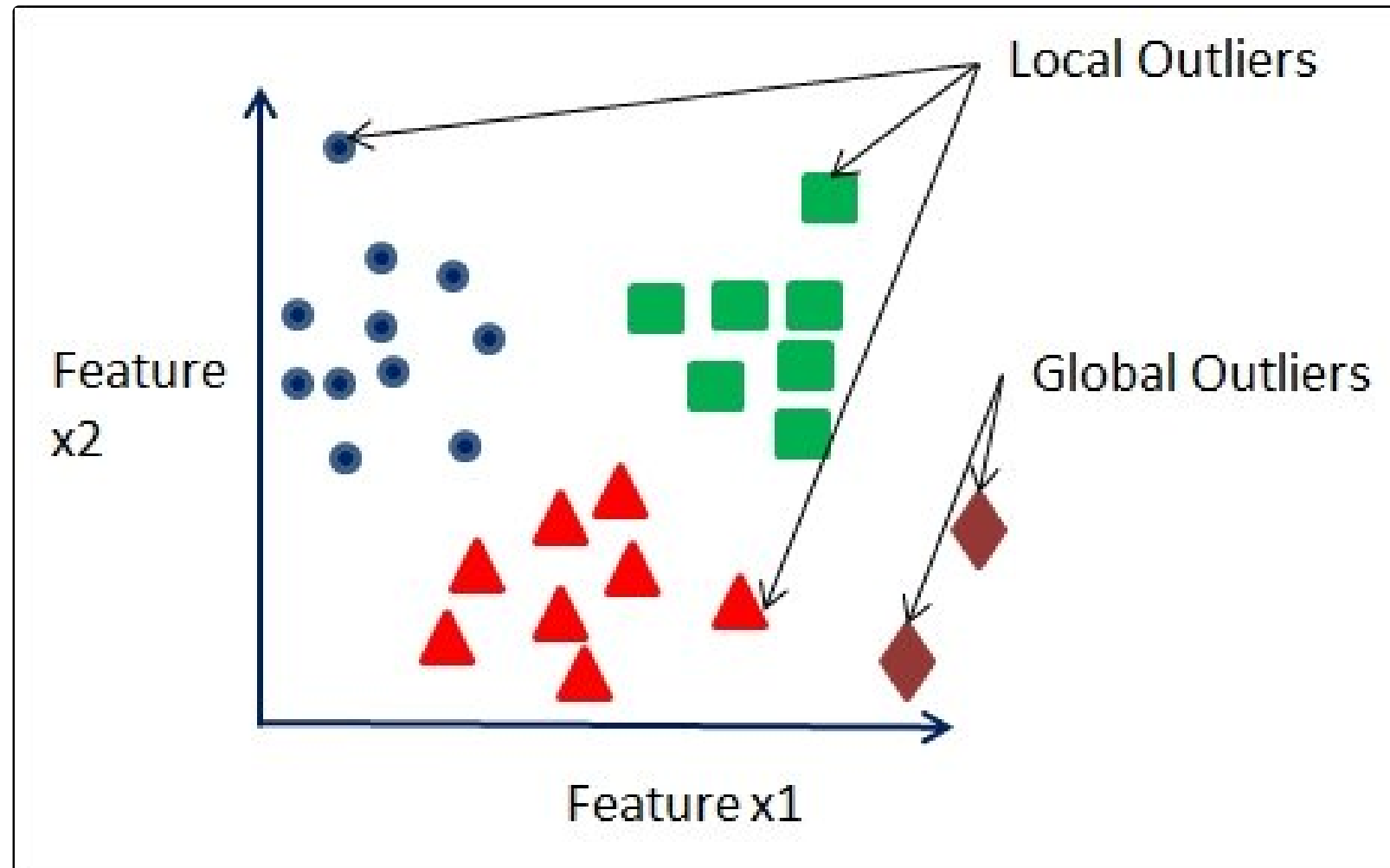
2026 Pre-Doctoral Economics Conference at Yale
Tobin Center for Economic Policy

The Problem

- Modern causal methods (DML, causal forests) use ML to predict **nuisance functions** $E[Y|X]$ and $E[D|X]$. Outliers distort these predictions and bias β
- Traditional fix (trim each column at 95th percentile) is **univariate** and removes **35–98%** of high-dimensional data
- With extreme events (COVID, tariffs, supply shocks) increasingly common, this can **flip the sign** of your treatment effect

$$Y = D\beta + g(X) + \varepsilon \implies \tilde{Y} = \tilde{D}\beta + \tilde{\varepsilon}$$

This Paper



- Evaluate 6 **multivariate, unsupervised** ML outlier methods from PyOD:
IForest, KNN, COPOD, ECOD, CBLOF, PCA
- Detect both **local** (between clusters) and **global** (extreme) outliers
- Propose an **ensemble**: only remove observations *all models agree* are outliers

Simulated Data: Can We Recover the Truth?

100K obs × 100 features | True ATE = **17.43** | True outlier % = 1%

Model	Data Removed	ATE	95% CI
No removal	0%	-0.01	[-0.02, 0.00]
Ensemble	1%	17.55	[16.68, 18.42]
Percentile 95th	98%	28.47	[18.93, 38.01]

1% outliers drove the ATE from 17.43 → 0. The ensemble recovers the truth.

Real Data: Does It Change Conclusions?

FICO HELOC data | **Y:** External Risk Score | **D:** Long Credit History (top 20%)

Model	Data Removed	ATE	95% CI	Significant?
No removal	0%	4.44	[0.53, 8.35]	Yes
Ensemble	0.8%	1.25	[−0.79, 3.28]	No
Percentile 95th	35.8%	0.26	[−1.70, 2.22]	No

A "significant" effect of 4.44 disappears, removing just 0.8% of data.

Takeaways

- A small % of outliers can **completely distort** causal estimates, even flipping signs
- Use an **ensemble with a high assumed outlier percentage**. It self-corrects and minimizes data loss
- Percentile trimming is **inadequate** for high-dimensional data. Multivariate methods dominate

Thank you! Questions?